Practice workbook · Complete a truth table

Filling in input combinations and intermediate and final columns for expressions using AND, OR and NOT.

5 question parts: 1 worked-example retry and 4 further practice parts.
Name: _____ Class: _____ Date: _____

Practise one question type at a time. Attempt the part inside the red boundary; the rest of the question and any figures are kept for context. Write your working in the answer space.

The first question is a retry of the website worked example, without its solution. Check your answers online with 'Check my answer', or ask your teacher for the answer booklet.

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Type ID: CSC-T4-10

1. Question 4 · part c.i · 3 marks

2025 JUN · Paper 01

Worked-example retry · Attempt part c.i only.

4 A house has solar panels and a battery storage system.

(a) A smartphone application is used to monitor the battery storage system remotely. The smartphone uses a 5G connection.

(i) Give **two** benefits of a 5G connection.

(2)

1 .....

.....

2 .....

.....

(ii) The application requires users to authenticate their identity.

One method of authentication is a user ID and password.

Describe **one other** method the application could use.

(2)

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(iii) Software developers perform code reviews.

Give **two** reasons for carrying out a code review.

(2)

1 .....

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2 .....

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1. Question 4 · part c.i · 3 marks

2025 JUN · Paper 01

Worked-example retry · Attempt part c.i only.

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(b) The battery storage system stores readings as signed 8-bit binary integers in **sign and magnitude** format.

Complete the table to show the 8-bit binary pattern and denary value for the lowest and the highest values that can be stored.

(2)

	8-bit binary pattern	Denary value
Lowest value		
Highest value		

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1. Question 4 · part c.i · 3 marks

2025 JUN · Paper 01

Worked-example retry · Attempt part c.i only.

(c) The solar panels and battery storage system are controlled by a program.

(i) An LED is displayed when the system is on (**O**), the sun is shining (**S**), and the house is not using all the power (**P**) generated by the solar panels.

Complete the truth table to show the results of each operation.

(3)

O	S	P	O AND S	NOT P	(O AND S) AND (NOT P)
0	0	0			
0	0	1			
0	1	0			
0	1	1			
1	0	0			
1	0	1			
1	1	0			
1	1	1			

(ii) The start-up instructions for the solar panels and battery storage system are stored in read-only memory (ROM).

Explain why the start-up instructions are stored in ROM.

(2)

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(Total for Question 4 = 13 marks)

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1. Question 4 · part c.i · 3 marks · Working and answer

Working and answer

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2. Question 5 · part a · 3 marks

2020 NOV · Paper 01

Further practice · Attempt part a only.

5 Programmers work as part of a team to develop software.

(a) They use truth tables to determine if an expression evaluates to true or false.

Complete this truth table for the Boolean expression

P AND (Q OR R)

You may not need to use all rows.

(3)

P	Q	R	Q OR R	P AND (Q OR R)

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2. Question 5 · part a · 3 marks

2020 NOV · Paper 01

Further practice · Attempt part a only.

(b) The team is working on software for a ride-on lawn mower.

The lawn mower must not start unless it is safe for the driver.

- The brake (B) must be engaged.
- The driver's seat sensor (S) must be engaged.
- The blade safety lock (L) must be removed.

Construct a Boolean expression, using AND, OR, and NOT with the letters B, S and L to show the safe starting condition.

(2)

(c) The team uses code reviews as part of the software development process.

State **two** ways in which code reviews can help protect software systems from cyber attacks.

(2)

1 .....

2 .....

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2. Question 5 · part a · 3 marks · Working and answer

Working and answer

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3. Question 4 · part c · 3 marks

2021 JUN · Paper 01

Further practice · Attempt part c only.

4 Scientists often use models and simulations.

(a) Give **one** reason, other than cost, why scientists use simulations.

(1)

.....

.....

(b) Explain **one** reason why a simulation may require a powerful processor.

(2)

.....

.....

.....

.....

(c) A scientist uses Boolean logic in programs.

Complete the table to show the results of each operation.

(3)

R	S	W	NOT S	R AND W	(NOT S) OR (R AND W)
0	0	0			
0	0	1			
0	1	0			
0	1	1			
1	0	0			
1	0	1			
1	1	0			
1	1	1			

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3. Question 4 · part c · 3 marks

2021 JUN · Paper 01

Further practice · Attempt part c only.

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(d) Von Neumann developed the stored program concept.

(i) Describe the **stored program concept**.

(2)

(ii) The fetch-decode-execute cycle is the cycle the central processing unit (CPU) follows in order to process instructions.

Name **two** registers used in the cycle.

(2)

1 .....

2 .....

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(iii) Describe the role of the address bus and the data bus during the **fetch** part of the cycle.

(2)

(iv) Complete the table to show the effect on a computer system of increasing the width of a bus.

(2)

Change made	Effect
Increasing the width of the data bus	
Increasing the width of the address bus	

(Total for Question 4 = 14 marks)

3. Question 4 · part c · 3 marks · Working and answer

Working and answer

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4. Question 5 · part d.i · 3 marks

2023 JUN · Paper 01

Further practice · Attempt part d.i only.

5 Hardware devices execute programs to carry out a variety of tasks.

(a) A program controls a bee character in an animation.

The bee can turn to face North, East, South or West.

The bee can move any number of steps in the direction it is facing.

Complete the table to show one input and one output required to move the bee.

(2)

Requirement	Example
Input	
Process	Calculate the path the bee will move along to its new position
Output	

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4. Question 5 · part d.i · 3 marks

2023 JUN · Paper 01

Further practice · Attempt part d.i only.

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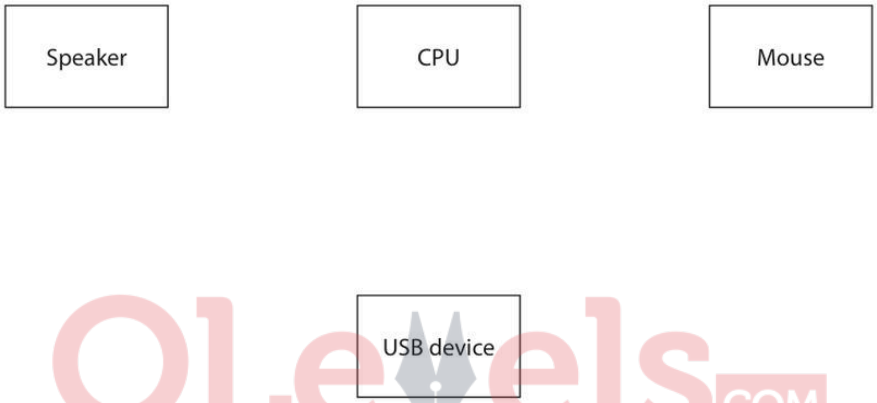
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(b) Computers are made up of hardware components.

(i) Complete the diagram by adding directional arrows between the components to show the flow of communication.

(3)



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4. Question 5 · part d.i · 3 marks

2023 JUN · Paper 01

Further practice · Attempt part d.i only.

(ii) Identify what is stored in ROM.

(1)

- ☒ **A** The software firewall
- ☒ **B** The basic input output system
- ☒ **C** The operating system
- ☒ **D** The user interface code

(iii) Cache is used as temporary storage.

One type of cache is located between main memory and the CPU.

Explain **one** reason cache is used in a computer.

(2)

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4. Question 5 · part d.i · 3 marks

2023 JUN · Paper 01

Further practice · Attempt part d.i only.

(c) A washing machine uses several different embedded systems.

One embedded system uses a switch to identify the type of wash cycle selected by the user.

Describe **one other** example of an embedded system found in a washing machine.

(2)

(d) Programs use logic statements to control physical hardware.

(i) A window shuts when the temperature is too cool or it is a rainy night.

The values are defined as:

- A shows it is night time
- B shows it is too cool
- C shows it is raining.

Complete the truth table to show the results of each operation.

Two rows have been done for you.

(3)

A	B	C	A AND C	(A AND C) OR B
0	0	0	0	0
0	0	1		
0	1	0		
0	1	1		
1	0	0		
1	0	1		
1	1	0		
1	1	1	1	1

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4. Question 5 · part d.i · 3 marks

2023 JUN · Paper 01

Further practice · Attempt part d.i only.

(ii) A warehouse has an automated alarm system.

When the alarm system is activated it will sound if:

- a movement sensor (M) is activated
- a pressure pad (P) is activated
- a key code (C) to deactivate the alarm system has not been entered.

Construct a logic statement, using AND, OR and NOT with the letters M, P and C, to show the conditions that will sound the alarm.

(3)

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4. Question 5 · part d.i · 3 marks · Working and answer

Working and answer

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5. Question 3 · part a · 4 marks

2024 JUN · Paper 02

Further practice · Attempt part a only.

3 Programs use logic, display data and validate input.

(a) The truth table shows the initial values for A and B.

Complete the truth table to show the results of each operation.

(4)

A	B	A OR B	NOT B	(A OR B) AND (NOT B)
0	0			
0	1			
1	0			
1	1			

(b) **Figure 2** shows the intended output from a program that displays every other name in an array of star names.

```
Alasia
Castor
Electra
Gudja
Izar
Kang
Maia
Ogma
```

Figure 2

Open **Q03b** in the code editor.

Amend the code to display every other name in the array of star names.

Do not add any further functionality.

Save your code as **Q03bFINISHED** with the correct file extension for the programming language.

(3)

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6

P 7 5 7 3 7 A 0 6 1 6

5. Question 3 · part a · 4 marks · Working and answer

Working and answer

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